

Predicting Overweight and Obesity in Later Life from Childhood Data: A Review of Predictive Modeling Approaches



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Abstract Overweight and obesity are an increasing phenomenon worldwide. Reliable and accurate prediction of future overweight or obesity early in the childhood could enable effective interventions by experts. While a lot of research has been done using explanatory modeling methods, capability of machine learning, and predictive modeling, in particular, remain mainly unexplored. In predictive modeling, the models are validated with previously unseen examples, giving a more accurate estimate of their performance and generalization ability in real-life scenarios. Our objective was to find and review existing overweight or obesity research from the perspectives of childhood data and predictive modeling. Thirteen research articles and three review articles were identified as relevant for this review. In general, prediction models with high performance either have a short time span to predict and/or are based on late childhood data. Logistic regression is currently the most often used method in forming the prediction models, although recently more complex models have also been applied. In addition to child's own weight and height information, maternal weight status and body mass index were often used as predictors in the models. More recent research has started to focus on a wider variety of other predictors as well.

Keywords Predictive models · Machine learning · Artificial intelligence · Obesity · Overweight

1 Introduction

Obesity is a global phenomenon that has increased rapidly during the last few decades in most countries. This trend has also led to significant increase in obesity-related diseases and deaths [15]. The review presented here aims to map and review the

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methods used in overweight and obesity prediction research with an emphasis on predictive modeling techniques. The basic question to consider is “Can we predict a person’s overweight or obesity status in later life from the data collected during childhood?” Ideally, early identification will make it possible to take steps for a successful obesity intervention. Currently, this identification is often done manually by using growth references such as de Onis et al. [12], Saari et al. [26], and Cole and Lobstein [9]. Also, the data collection phase can be a tedious and expensive process. It is therefore preferred that this identification can be achieved with easily available basic data, such as height and weight information. A widely used measure derived from these two attributes is body mass index (BMI). Ideally, children in unhealthy BMI trajectories should be identified before school age [21]. For adults, the BMI cut-off points in widest use are 25 kg/m^2 for overweight and 30 kg/m^2 for obesity [11]. Age and sex specific cut-off values that can be used in children have also been developed [10, 11].

When validating the employed prediction model, the performance measures are always a tradeoff between sensitivity (in this case the ratio of correctly classified overweight or obese cases in relation to total overweight or obese cases) and specificity (the ratio of correctly classified normal weight cases in relation to total normal weight cases). In this context, the most important performance measure is sensitivity, which indicates the proportion of children we are most interested in finding, in order to target preventive actions in this group [34]. However, when the specificity is low, Butler et al. [5] argue that it becomes questionable to use the model at all, since the model generates large amounts of false positives. They argue further that the risk threshold should be placed based on considering multiple criteria, including potential risks and harms as well as financial costs. Additionally, if a suitable tool for obesity prediction became eventually available for clinical use, the practitioners should take careful steps to avoid potential ethical issues in interventions. Communicating the overweight/obesity risk to child’s parents might have undesirable and unforeseen consequences. Also, availability of practical remedies should be ensured for those deemed to be at significant risk for obesity [20].

Breiman [4] and Shmueli [30] have discussed the differences between explanatory and predictive types of modeling. Explanatory modeling is defined as “the use of statistical models for testing causal explanations” and predictive modeling as “the process of applying a statistical model or data mining algorithm to data for the purpose of predicting new or future observations” [30]. Validation is used in predictive modeling to estimate how well the model is expected to work with previously unseen data [3]. According to Bzdok et al. [6], prediction makes it possible to identify the best courses of action without requirement to understand the underlying mechanisms. In explanatory modeling, the model is not validated with an independent test data and thus its performance in the task of predicting new observations might be overestimated [28].

2 Materials and Methods

The scope of this review included studies in English language that

1. Predict the future overweight/obesity status with a model built using the baseline data collected in childhood (e.g., using the data collected up to the age of three years to predict child's obesity status at the age of six years). This criterion includes only studies that either build binary classifiers or somehow employ regression in solving the classifier problem.
2. In addition to describing the prediction framework also report relevant numeric results (e.g., sensitivity and specificity and/or AUC values).
3. Validate the results using an internal and/or external independent test set that has not been used for training the model. Internal bootstrap validation is not considered here to be an independent test data set.

Exceptions to these criteria were made for existing surveys, literature reviews, and meta-analyses directly concerned in a similar prediction problem. An overview of the study selection process is presented in Fig. 1. Studies fulfilling the criteria were searched initially from two databases, PubMed¹ and IEEE Xplore,² in addition to extensive searches on Google Scholar.³ In the initial search phase, potential studies were mapped based on the titles of studies. PubMed search term was

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((bmi[Title] OR (body mass index[Title])) AND (obesity[Title] OR obese[Title]  
OR overweight[Title]) AND (prediction[Title/Abstract] OR predicting[Title/  
Abstract])).
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This search yielded 63 results in total, of which four were identified as potential for inclusion. For IEEE Xplore, “obesity” search term was used with conference articles, journals, and magazines starting from 1994 included in the search. The search yielded 634 results with two potential studies. An additional set of potential studies were then searched using Google Scholar, by using variations of the search terms described before. These initial searches were made in July 2017. After this initial seed set of potential studies was identified, references in studies and newer research that cite the studies were mapped for finding additional potentially relevant studies. Some of the relevant studies were identified already during these steps.

A second search phase is indicated by a two-way arrow in Fig. 1. This searching through references of the potential studies and research that cited the potential studies continued until no new potential studies emerged. Each search cycle produced new material to the pool of potentially relevant studies. The idea and assumption behind

¹ <https://www.ncbi.nlm.nih.gov>.

² <http://ieeexplore.ieee.org>.

³ <https://scholar.google.com>.

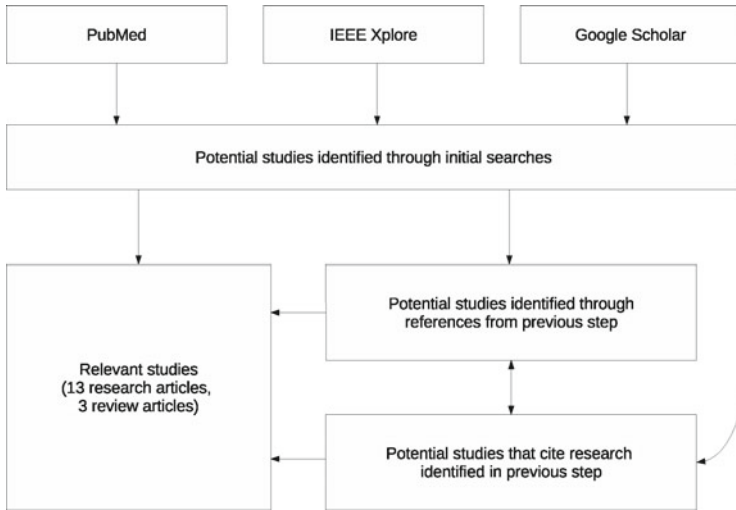


Fig. 1 An overview of the study selection process

this approach was that the relevant studies are most likely referenced in recent (2010–2018) research articles. Hence the search focused on the references of recent studies that were identified as potential for inclusion. This search phase lasted until August 2018. To find the most recent studies released after 2018, the second search phase was repeated in July 2020. During the final search, five new articles released at the latest in June 2020 were identified and added to the set of included studies.

3 Results

To our knowledge, this is the first study collecting the existing research that focuses on the predictive modeling approach in prediction of overweight or obesity. Three surveys, literature reviews and meta-analyses from the years 2010 to 2015 closely related to the problem of overweight/obesity prediction are presented and discussed in Sect. 3.1.

Eight out of 13 studies from 2004 to 2020 presented in Sect. 3.2 employ logistic regression in their models. Only two early studies [14, 34] explore more complex models such as decision trees, Bayesian and neural networks, support vector machines, that have greater capacity to learn associations from data. Recently, in 2019 and 2020, more complex models [17, 18, 24, 32] have been explored more extensively. Seven studies predict only obesity and five only overweight including obesity. One study [7] predicts separately two cases: overweight (including obesity) and obesity. The age range to predict overweight/obesity status in the studies was from two years to 33 years, with ten studies predicting status in children under ten

years of age. In addition to weight and height information of the child, number of features used in prediction varied from zero to over 1700. Generally, more complex models employed recently used more features than simpler models presented in the earlier studies.

3.1 Surveys, Literature Reviews, and Meta-analyses

A survey of data mining methods used in the field was conducted by Adnan et al. [1]. The study summarized some of the research in the area and focused on describing three methods used in three different studies: neural networks, naïve Bayes classifiers, and decision trees. Each of the methods was described as having its own strengths and weaknesses. The conclusion was that to improve prediction results, further improvements of the techniques are necessary. The authors planned to continue their work by combining different existing methods to form a single better performing hybrid method.

Infancy weight gain was used as a predictor for childhood obesity in a meta-analysis presented by Druet et al. [13]. The meta-analysis reported a consistent positive association of infancy weight gain to subsequent obesity. The study included ten cohort studies from which full data of three cohorts ($N = 8236$) were used in forming models for overweight and obesity prediction. A second sample of the same size was used for validating the results. The prediction model employed stepwise multivariable logistic regression, and used mother's BMI, child's birthweight and sex in addition to weight gain information from birth to one year of age for childhood obesity prediction. The model was reported to show moderate predictive ability with area under curve (AUC) value of 0.77. When using a risk score threshold that puts 10% of the population above the threshold, the model had a sensitivity of 58.6% and specificity of 90.9%. A similar model was also created for childhood overweight prediction and reported AUC value was 0.76.

An extensive systematic review and meta-analysis of existing studies were presented by Simmonds et al. [31] to examine the use of different measures of obesity in childhood for predicting obesity and development of obesity-related diseases. The study found that the predictive accuracy of childhood obesity for predicting adult obesity had a sensitivity of 30% and a specificity of 98% and was described as moderate. The study concluded that childhood BMI is not an effective predictor for obesity or disease in adulthood, since most obese adults were not obese in childhood. However, no evidence was found in the study to support any other single measure over BMI.

3.2 Predictive Modeling Approaches

An overview of the studies employing predictive modeling is presented in Table 1. The studies might present also models that have not been validated with an independent data set, i.e., be partly explanatory in their approach. *Predicted status* (overweight/obesity) as well as other information is listed only if the results were validated independently. The number of *additional features* used exclude sex, weight, height, BMI of the child and overweight/obesity labels. If the study had separate models for females and males, best results (AUC, sensitivity, and specificity) are presented for both. Two of the studies [16, 34] presented different models for predicting overweight based on data collected up to three points of time. The best result for each of these models is also presented in the tables. In addition to these, Cheung et al. [7] presented separate models for obesity and overweight prediction. Again, the best results for both models are presented.

Data collected at childhood was used by Cheung et al. [7] to predict obesity and overweight at the age of 33 years and self-reported disease history at the age of 42 years. ROC analysis was used to define optimal BMI risk thresholds at the ages of seven, 11, and 16 years. The cohort data was split in half, with the first half used for the ROC analysis and the second one for validating the cut-off BMI thresholds. For the validation data ($N = 4231$) at the age of 11 years, sensitivity and specificity values for obesity status prediction were, respectively, 71.7% and 72.4% for males. For overweight prediction, the same values were 65.6% and 68.6%. Obesity prediction for females achieved sensitivity of 75.7% and specificity of 69.7%, while the same values for overweight prediction were 69.8% and 63.6%.

Six independent data mining methods were investigated and compared with the conventional logistic regression approach by Zhang et al. [34]. The study included C4.5 decision tree, Bayesian networks, naïve Bayes, association rules, neural networks, and support vector machines (SVMs). The task was to predict overweight status at the age of three years based on child's data ($N = 16523$) collected up to six weeks, eight months, and two years, respectively. 67% of the data was used for training the models and 33% for testing. Children's height and weight data were used in the models. Time of gestation was used as an additional feature when predicting overweight status from the age of two years. The results showed that the prediction accuracy improved with the methods used when compared with the logistic regression approach. The best performing algorithms were SVM with radial basis function (RBF) kernel and Bayesian methods, specifically naïve Bayes. At the age of two years, the sensitivity and specificity of NB were 54.7% and 93.1%, respectively, while the same values for SVM were 60.0% and 79.6%. The conclusion was that to improve overweight prediction rates more features may need to be recorded and used for prediction.

The aim of Dugan et al. [14] was to improve the work presented in Zhang et al. [34] by considering a significantly extended set of predictors. A partly different set of machine learning algorithms were also explored in the study. They included random tree, random forest, ID3, and C4.5 decision trees in addition to naïve Bayes and

Table 1 Overview of the studies using predictive modeling methods for overweight/obesity prediction based on childhood data

Study	Method(s)	Predicted status	Age(s) to predict	Predicts status from child's data recorded up to	Weight and height information used	Number of additional features used	Additional features	Validation AUC (best)	Validation sensitivity (best)	Validation specificity (best)	N (training data)	N (validation/testing data)
Cheung et al. [7]	ROC threshold cut-off	Overweight (incl. obesity) and obesity	33 years	11 years	BMI	0	N/a	N/a	75.7%*, 71.7%** (obesity), 69.8%*, 65.6%** (overweight)	69.7%*, 72.4%** (obesity), 63.6%*, 68.6%** (overweight)	4136	4231
Zhang et al. [34]	Logistic regression, decision tree, association rules, neural network, linear SVM, RBF SVM, Bayesian network, naïve Bayes	Overweight (incl. obesity)	3 years	6 weeks, 8 months, 2 years	Birth weight, weight change, height, BMI	0 (6 weeks), 0 (8 months), 1 (2 years)	Time of gestation	N/a	11.2% (6 weeks), 35.5% (8 months), 54.7% (2 years)	96.0% (6 weeks), 91.5% (8 months), 93.1% (2 years)	≈11070 ^g	≈5400–5600 ^h
Dugan et al. [14]	Random tree, random forest, ID3 decision tree, J48 decision tree, naïve Bayes, Bayesian network	Obesity	2–10 years ^a	<2 years	Height and weight	Over 160	Multiple ^d	N/a	89%	83%	≈6767 per fold (10-fold cv)	≈752 per fold (10-fold cv)

(continued)

Table 1 (continued)

Study	Method(s)	Predicted status	Age(s) to predict	Predicts status from child's data recorded up to	Weight and height information used	Number of additional features used	Additional features	Validation AUC (best)	Validation sensitivity (best)	Validation specificity (best)	N (training data)	N (validation/ testing data)
Morandi et al. [22]	Stepwise logistic regression	Obesity	7 years	Birth	Birth weight	5	Maternal BMI, paternal BMI, number of household members, maternal occupation, gestational smoking	0.73	N/a	N/a	4032	1503 and 1032
Santorelli et al. [29]	Stepwise logistic regression	Obesity	2 years	9 months, 12 months	Birth weight and weight change	0	N/a	0.850 (9 months), 0.886 (12 months)	N/a	N/a	1528 (9 months), 731 (12 months)	880 (9 months) 867 (12 months)
Weng et al. [33]	Stepwise logistic regression	Overweight (incl. obesity)	3 years	1 year	Birth weight and weight change	4	Maternal pre-pregnancy weight status, paternal BMI, maternal smoking during pregnancy, breastfeeding in the first year	0.755	76.9%	66.5%	8299	1715

(continued)

Table 1 (continued)

Study	Method(s)	Predicted status	Age(s) to predict	Predicts status from child's data recorded up to	Weight and height information used	Number of additional features used	Additional features	Validation AUC (best)	Validation sensitivity (best)	Validation specificity (best)	N (training data)	N (validation/testing data)
Redsell et al. [25]	Stepwise logistic regression	Overweight (incl. obesity)	5 years	1 year	Birth weight and weight change	4	Maternal pre-pregnancy weight status, paternal BMI, maternal smoking during pregnancy, breast-feeding in the first year	0.79	53%	71%	8299	980
Gravensen et al. [16]	Logistic regression	Overweight (incl. obesity)	13–16 years ^b	Birth, 5 years, 8 years	Birth weight and BMI	1	Maternal BMI	N/a	24.0%*, 17.4%** (birth), 38.9%*, 28.2%** (5 years), 49.2%*, 38.7%** (8 years)	92.1%*, 91.7%** (birth), 94.4%*, 94.2%** (5 years), 96.0%*, 96.7%** (8 years)	4111	5414
Hammond et al. [18]	Logistic regression, LASSO regression, random forest classifier and regression, gradient boosting classifier and regression	Obesity	5 years	2 years	Weight-for-length and BMI	1509 in total ^c	Multiple ^e	0.817*, 0.761**	0.491* ^f , 0.699** ^f	0.912* ^f , 0.670** ^f	2760	689

(continued)

Table 1 (continued)

Study	Method(s)	Predicted status	Age(s) to predict	Predicts child's data recorded up to	Weight and height information used	Number of additional features used	Additional features	Validation AUC (best)	Validation sensitivity (best)	Validation specificity (best)	N (training data)	N (validation/testing data)
Saïjo et al. [27]	Logistic regression	Obesity	6–8 years	1 year	Obesity index at one year of age	3	Information about the mother: pre-pregnancy BMI, smoking during pregnancy, education	0.688	49.7%	83.8%	5488	1358
Pang et al. [24]	XGBoost	Obesity	2–7 years	2 years	Weight, height, BMI-for-age and other derived information	102 in total	Multiple features incl. demographic features and laboratory values	0.81	79.91%	63.27%	≈21762	≈5441
Gupta et al. [17]	Recurrent neural network (RNN) with LSTM	Obesity	All ages from 3 up to 20 years	Multiple observation windows employ data from age of 0 to 17 years	At least weight, height, BMI and other values generated from the data	1737 in total	Thousands of variables derived from various EHR data	One-year: 0.87–0.97, two-year: 0.82–0.96, three-year: 0.80–0.95 (estimated from the plots)	One-year: 0.63–0.91, two-year: 0.66–0.92, three-year: 0.63–0.87	N/a	≈40817	≈13606

(continued)

Table 1 (continued)

Study	Method(s)	Predicted status	Age(s) to predict	Predicts status from child's data recorded up to	Weight and height information used	Number of additional features used	Additional features	Validation AUC (best)	Validation sensitivity (best)	Validation specificity (best)	N (training data)	N (validation/testing data)
Singh and Tawfik [32]	kNN, J48 pruned tree, random forest bagging, SVM, multilayer perceptron (MLP), voting	Overweight (incl. obesity)	14 years	11 years	BMI	0	N/a	N/a	62–94% (depending on the setting)	N/a	≈7777	≈3333

*Females
**Males
^aIf the child was obese at any time between the ages of two and ten years, he/she was labeled as obese
^bIf the child was overweight at any time between the ages of 13 and 16 years, he/she was labeled as overweight
^cOriginal feature space of 19290 features shrunk to 8% when a minimum of five children with information was used as a threshold
^d167 variables in total, but they are not listed in detail. A questionnaire form used to collect the information is presented
^eAvailable features are listed using 23 main categories
^fThe best model was presented as a performance tradeoff table. The values presented here are taken from the point of the highest F1 score value
^g67% of total data
^h33% of total data from 16523 equals 5453 observations. However, the confusion matrices in Figs. 3 and 4 in the study include 5618 observations

Bayesian networks. The study employed in total 167 predictors that were collected through questionnaires filled by parents and physicians. The data ($N = 7519$) were collected before the children's second birthday. 10-fold cross-validation was used for validating the results. If the child was obese at any point after her/his second birthday to the age of 10 years, the child was labeled as obese. The prediction task consisted of predicting the child's future obesity status after the second birthday. Two of the algorithms, ID3 and NB, performed slightly better when predictors described as "noisy" were removed from the data. The ID3 model used 87 and the NB model 107 out of 167 predictors. Other methods employed in the study performed best when all the available predictors were utilized. The study singled out two algorithms as best working with this data set: ID3 decision tree with sensitivity and specificity of 89% and 83%, respectively, along with random tree's performance metrics of 88% and 80%.

Stepwise logistic regression analysis was used to form predictive models by Morandi et al. [22] to predict obesity status at the age of seven years with data available at birth. The final model included maternal and paternal BMI, number of household members, maternal occupation, gestational smoking as well as birth weight. First cohort data ($N = 4032$) was used in training the model, while validation of the results was performed using two independent cohorts ($N_1 = 1503$, $N_2 = 1032$). Reported AUC values for the two validation sets were 0.70 and 0.73 respectively. For the first validation cohort, gestational smoking and number of household members information had to be omitted from the model. The study also experimented with adding a genetic score information to the models, but that did not provide any significant improvement in terms of predictive power.

Three logistic regression models were formed with stepwise method by Santorelli et al. [29], where the models were applied in a mobile application aimed at parents. The models predicted the risk of childhood obesity at the age of two years. The plan presented in the study was that the parents use the application when their infant's age reaches six, nine, and twelve months, with separate models presented for each case. Input features included sex, birth weight, and weight gain. Two datasets were used. The first one was used to form the three models ($N_1 = 1022$, $N_2 = 1528$, $N_3 = 731$) and validated using an internal bootstrap validation. The second dataset was used as a separate external testing data. The second and third models were validated using this data ($N_2 = 880$, $N_3 = 867$), with the reported AUC values for the two models being 0.85 and 0.89, respectively. More detailed sensitivity and specificity values were also reported for different model configurations, but only for the training data. The mobile application has since been discontinued, and no published research exists on the usage or effectiveness of the application [5].

Weng et al. [33] formed a childhood overweight prediction model (IROC) using stepwise logistic regression for predicting obesity status at the age of three years. The cohort data was randomly divided into training ($N = 8299$) and testing ($N = 1715$) sets. The data included 33 potential predictor variables. From these, seven significant input features were identified: sex, birth weight, weight gain in first year, maternal pre-pregnancy weight status, paternal BMI, maternal smoking during pregnancy and breastfeeding in the first year. The study reported a moderately good predictive

ability, with sensitivity value of 0.769, specificity of 0.665 and AUC value 0.755 for the testing data.

The IROC algorithm [33] was further validated by Redsell et al. [25] with an additional independent dataset ($N = 980$), predicting overweight status at the age of five years. Four models were formed for prediction. The first one (clinical model) used the original algorithm directly and assigned null values to missing data. The reported AUC values were 0.67 when using the International obesity taskforce overweight criteria [11] and 0.65 when using the UK 1990 overweight criteria [10]. The second one (recalibrated model) used multivariate logistic regression to generate and recalibrate the model to reflect the demographics of the new validation data (AUC values of 0.70 and 0.67 were reported). The third one (imputed model) used multiple imputation to generate ten copies of the existing data set to predict missing risk factor from multivariate models (AUC values 0.79 and 0.73). The fourth one (recalibrated imputed model) applied the recalibrated algorithm to the imputed data (AUC values 0.93 and 0.90).

Graversen et al. [16] developed logistic regression models to predict adolescent overweight, adult overweight, and adult obesity. Input features for the models included maternal BMI, birth weight, and early childhood BMI. First dataset ($N = 4111$) was used to form the models. Performance of the models was validated with an internal bootstrap validation as well as with an external independent dataset ($N = 5414$), where the prevalence of overweight was much higher. Only the model for adolescent overweight prediction was validated with the external data. The study reported results for adolescent overweight prediction from data collected at birth, up to the age of five years, and up to the age of eight years. Also, different thresholds of percentage of children labeled as “at risk” of overweight were explored. With the external validation dataset, sensitivity and specificity for adolescent overweight prediction at the age of five years for females were 38.9% and 94.4%, respectively, when the threshold of being at risk was set to upper 10%. For males these measures were 28.2% and 94.2%.

Hammond et al. [18] employed a set of classifier and regression models for predicting the obesity status at the age of five years. The regression models were configured to predict the BMI value, which was then converted to a binary value indicating if the child was predicted to be obese or not. The models were formed separately for girls and boys. Their initial set ($N = 3449$) of predictors was very large at 19,290 features, but shrunk down to 8% when a minimum of five children with information was used as a threshold. For validation, the study held out a random test set that included 20% of the data. The remaining data was then used for training the model with an internal bootstrap validation. The best performing models with the highest mean test data AUC values (0.817 for girls and 0.761 for boys) were achieved with LASSO regression for both sexes. The final LASSO regression model included 36 features for girls and 50 features for boys. The features associated strongest with obesity for both sexes were weight-for-length z-score and BMI.

Saijo et al. [27] divided a total sample of mother-child pairs ($N = 6846$) randomly into derivation (80%) and validation (20%) cohorts. The study identified a set of obesity predictors based on univariate and multivariate logistic regression analysis

employing data from pregnant women and one year old infants. The predictors identified in the model derivation phase included mother's pre-pregnancy BMI, child's gender, smoking during pregnancy, education, and obesity index at the age of one year. The predicted outcome was the obesity index of over 20% from the weight-for-height charts for Japanese children at the age of 6–8 years. The reported AUC for the validation cohort was 0.688, with sensitivity of 68.8% and specificity of 83.8%.

Pang et al. [24] employed the gradient boosting library XGBoost for their obesity prediction model. The target was to predict, based on the child's data collected up to the age of two years, if the child was to be obese at any stage between the ages of two and seven years. The data ($N = 27203$) was divided into two separate training sets (40% each) and a test set (20%). The training sets were utilized in hyperparameter optimization conducted using 10-fold cross-validation. The study highlighted two new predictive features related to obesity: body temperature and respiratory rate. Other predictive features reported were weight, height, race, and ethnicity. The model AUC for the test set at sensitivity level of 80% was 0.81 with specificity of 63.27%.

Gupta et al. [17] approached the problem of obesity prediction with deep learning, using recurrent neural network (RNN) architecture with long short-term memory (LSTM) cells. The models were configured to predict the BMI value. The data ($N = 68029$) consisted of over 44 million records. After cleaning the variables that did not have enough information, the final data consisted of 1737 variables in total. Thousands of variables available in the EHR system are employed to form the time-series data for the patients. The time-series data included thousands of variables derived from condition diagnosis, drug prescription data, information on performed procedures, and laboratory results. In addition, there were static variables indicating sex, race, ethnicity, and zip code. 10-fold cross-validation was employed, with 60% of the data used for training, 20% for validation and 20% for the final test set. Data from 16 two-year observation windows were used to predict obesity at prediction windows placed at one, two, and three years ahead of the observation window. In total, the prediction models were formed for 48 sub-cohorts. The prediction results varied between sub-cohorts. For one-year predictions reported AUC was between 0.87–0.97, for two-year predictions 0.82–0.96, and for three-year predictions 0.80–0.95 (when estimated from the reported plots). Additional measures reported included sensitivity, accuracy, and positive predictive value (e.g., precision). The three most important variables obtained by averaging the importance scores were BMI (percentile) per age and gender, obese/non-obese label, and allergic urticaria. Their work also aimed to add interpretability to the proposed model. Interpretability was of the results was also considered on two levels: time-level and variable-level.

Singh and Tawfik [32] compared seven different classifier methods in the overweight prediction domain. For each child in the study ($N = 11110$), BMI was recorded at the ages of 3, 5, 7, and 11 years. This information was used in the prediction of overweight status at the age of 14 years. The results were reported for two cases. In the first case, the models were trained using the original imbalanced data. The best-reported sensitivity, attained using the J48 pruned tree, in the “normal” class was 0.94, with “at risk” class sensitivity of 0.62. Corresponding precision values were 0.87 and 0.78. In the second case, an oversampling method was employed

to the minority class to balance the data. Training the models with the balanced data improved the sensitivity of the “at risk” class. The sensitivity for the “normal” class in the second case with the MLP method was 0.62, with “at risk” class sensitivity of 0.92. The corresponding precision values were 0.96 and 0.51.

4 Conclusions

This review explored the existing research on overweight and obesity prediction. While various explanatory models have been studied and employed extensively in the research area, utilization of predictive modeling methods of machine learning remains partly unexplored in the field.

The studies using explanatory modeling do not validate the formed models with a separate test data. Instead they examine how well the whole data fits to the model. Generalization refers to how well the model trained on the training data set predicts the output for new instances, and it is an integral part of the machine learning and predictive modeling approach [2, 3]. We argue that if the model is aimed for prediction, the model should always be validated with independent data to get more reliable performance estimates.

In terms of predictive power, best performing models in the study either made the prediction quite late or had a relatively short period between the prediction and the outcome. Graversen et al. [16] had a very high specificity (96%) in predicting overweight at adolescence in girls, using data recorded up to eight years. Other moderately successful models [14, 24, 29] and “two years to three years” model by Zhang et al. [34] only had a short time period to cover. Recent studies [17, 18], which have employed more data and models with more complexity, have shown some promise when making predictions over longer time periods. So far there has been no evidence of success in employing any of the presented or other models in clinical use [5].

The constantly growing size of data sets will enable the use of even more powerful machine learning methods and, moreover, sophisticated analysis of the obtained models. In the present context, one of the most promising approach is, for example, the recurrent neural networks [19], currently explored only in one study [17]. They could provide a powerful method for predicting overweight and obesity development in the later life, because they are inherently designed for time-series prediction tasks (e.g., [8]). It is, however, important to collect a sufficient amount of data before training complex models, such as neural networks.

Besides being powerful methodology for building prediction models, more effort could also be directed to machine learning-based hypothesis generation by employing large data sets, high-performance computing and machine learning algorithms. This approach can lead to finding of unsuspected information and predictors from the growing data.

A critical issue to be considered when applying machine learning in overweight and obesity prediction tasks is, however, the risk of chance findings. In order to

minimize the risk of chance models or predictors, it is highly important to develop and apply strategies, such as data randomization [23], for confirming significance of the obtained models and relevance of the identified predictors.

5 Summary Points

What was already known on the topic

- A lot of research has been done on predicting upcoming obesity or overweight.
- None of the prediction models presented in previous research have been successfully utilized in practice.

What this study added to our knowledge

- Although prediction in domain of overweight/obesity has been studied extensively, there are not many studies that separate the training and validation of the model in a way expected in predictive modeling.
- Highest performing prediction models are either only predicting near future overweight/obesity status or make their prediction relatively late.
- More complex models with greater learning capacity have been employed in only a limited set of studies. Future research possibilities in the area exist in examining the potential of recurrent models further and considering further model building practices such as predictor selection and significance of prediction.

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